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-- Part 1: Data Definition Language (DDL)

-- =====

-- 1. Database Creation

CREATE DATABASE IF NOT EXISTS `Pine ValleyFC`;

USE `Pine ValleyFC`;

-- 2. Create Tables (Parent Tables First)

CREATE TABLE Customer_T (

CustomerID INT PRIMARY KEY,

CustomerName VARCHAR(100) NOT NULL,

CustomerAddress VARCHAR(150),

CustomerCity VARCHAR(50),

CustomerState VARCHAR(50),

CustomerPostalCode VARCHAR(20)

);

CREATE TABLE Product_T (

ProductID INT PRIMARY KEY,

ProductDescription VARCHAR(100),

ProductFinish VARCHAR(50),

ProductStandardPrice DECIMAL(10,2),

CONSTRAINT chk_ProductFinish CHECK (ProductFinish IN ('Natural Ash', 'Red Oak', 'Cherry', 'Walnut'))

);

-- Child Tables (Referencing Parent Tables)

```
CREATE TABLE Order_T (  
    OrderID INT PRIMARY KEY,  
    OrderDate DATE,  
    CustomerID INT,  
    FOREIGN KEY (CustomerID) REFERENCES Customer_T(CustomerID) ON DELETE  
    RESTRICT ON UPDATE CASCADE  
);
```

```
CREATE TABLE OrderLine_T (  
    OrderID INT,  
    ProductID INT,  
    OrderedQuantity INT,  
    PRIMARY KEY (OrderID, ProductID),  
    FOREIGN KEY (OrderID) REFERENCES Order_T(OrderID) ON DELETE RESTRICT ON  
    UPDATE CASCADE,  
    FOREIGN KEY (ProductID) REFERENCES Product_T(ProductID) ON DELETE RESTRICT  
    ON UPDATE CASCADE  
);
```

-- 3. Schema Modification

```
ALTER TABLE Customer_T ADD CustomerEmail VARCHAR(50) UNIQUE;
```

-- 4. Create Index for performance

```
CREATE INDEX idx_product_price ON Product_T(ProductStandardPrice);
```

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-- Part 2: Data Manipulation Language (DML)

-- =====

-- 1. Insertion

INSERT INTO Customer_T (CustomerID, CustomerName, CustomerCity, CustomerState) VALUES

(1100, 'Evince', 'Accra', 'Greater Accra'),
(1101, 'Gabby', 'Kumasi', 'Ashanti'),
(1102, 'Theophilus', 'Koforidua', 'Eastern'),
(1103, 'George', 'Accra', 'Greater Accra'),
(1104, 'Ayoola', 'Tamale', 'Northern');

INSERT INTO Product_T (ProductID, ProductDescription, ProductFinish, ProductStandardPrice) VALUES

(1, 'Office Chair', 'Natural Ash', 150.00),
(2, 'Study Desk', 'Red Oak', 300.00),
(3, 'Bookshelf', 'Walnut', 450.00),
(4, 'Side Table', 'Natural Ash', 100.00),
(5, 'Modern Workstation', 'Red Oak', 1200.00),
(6, 'Cherry Dining Table', 'Cherry', 800.00);

INSERT INTO Order_T (OrderID, OrderDate, CustomerID) VALUES

(101, '2026-05-01', 1100),
(102, '2026-05-02', 1101),
(103, '2026-05-03', 1102),
(104, '2026-05-04', 1103),
(105, '2026-05-05', 1104);

INSERT INTO OrderLine_T (OrderID, ProductID, OrderedQuantity) VALUES

(101, 1, 12),
(102, 2, 5),

(103, 3, 15),

(104, 4, 2),

(105, 6, 1);

-- 2. Selection Queries

-- (a) Filtering & Sorting — FIXED: filter by city values, not state names

```
SELECT CustomerName, CustomerCity, CustomerState FROM Customer_T  
WHERE CustomerCity IN ('Accra', 'Kumasi') ORDER BY CustomerName ASC;
```

-- (b) Aggregate Functions

```
SELECT MAX(ProductStandardPrice) AS Max_Price,  
       MIN(ProductStandardPrice) AS Min_Price,  
       AVG(ProductStandardPrice) AS Avg_Price  
FROM Product_T;
```

-- (c) Inner Join

```
SELECT Order_T.OrderID, Customer_T.CustomerName FROM Order_T  
INNER JOIN Customer_T ON Order_T.CustomerID = Customer_T.CustomerID;
```

-- (d) Group By & Having

```
SELECT ProductID, SUM(OrderedQuantity) AS Total_Quantity FROM OrderLine_T  
GROUP BY ProductID HAVING SUM(OrderedQuantity) > 10;
```

-- 3. Update

```
UPDATE Customer_T SET CustomerCity = 'Koforidua' WHERE CustomerID = 1100;
```

-- 4. Delete

```
DELETE FROM Order_T WHERE OrderDate < '2023-01-01';
```

```
-- =====
```

```
-- Part 3: Data Control Language (DCL)
```

```
-- =====
```

```
CREATE USER IF NOT EXISTS 'sales_clerk'@'localhost' IDENTIFIED BY 'password123';
```

```
GRANT DELETE ON `Pine ValleyFC`.Product_T TO 'sales_clerk'@'localhost';
```

```
-- 1. Granting Permissions
```

```
GRANT SELECT, INSERT ON `Pine ValleyFC`.Order_T TO 'sales_clerk'@'localhost';
```

```
-- 2. Revoking Permissions
```

```
REVOKE DELETE ON `Pine ValleyFC`.Product_T FROM 'sales_clerk'@'localhost';
```

```
FLUSH PRIVILEGES;
```